Vel Tech
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R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of U.G.C. Act 1956)

Batch No:MP-AIDS-03

School of Computing

Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

Academic Year: 2025-2026

Major Project - Abstract Form

STUDENT DETAILS:

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TITLE & ABSTRACT:

Title: An AI-Based Approach for Personalized Clothing Recommendations Using Real-Time Emotion Recognition.

Abstract: Personalization in fashion has become a key area of innovation, aiming to enhance user satisfaction and engagement through intelligent systems. This project presents an AI-based approach that delivers personalized clothing recommendations by integrating real-time emotion recognition with individual user preference modeling. The system uses computer vision and deep learning techniques to detect facial expressions via webcam, identifying emotions such as happiness, sadness, anger, and surprise through convolutional neural networks (CNNs) trained on labeled emotion datasets. Detected emotional states are combined with historical clothing preferences and user profile data to generate context-aware suggestions using a hybrid recommendation engine that merges collaborative and content-based filtering techniques. By analyzing the user's current mood, the system suggests outfits that either complement or uplift their emotional state, contributing to a more engaging and psychologically aligned fashion experience. Real-time responsiveness is ensured through optimized CNN models and efficient facial detection algorithms, making it suitable for integration into various digital platforms. The recommendation engine adapts over time through user feedback, improving the relevance and personalization of suggestions. Experimental results on a custom dataset integrating emotion recognition and clothing choices show that the system offers enhanced recommendation quality over traditional methods. This study highlights the potential of emotion-aware AI in redefining fashion personalization and points to future directions such as multi-modal emotion sensing, integration with virtual try-on features, and adaptation to culturally diverse fashion preferences.

Keywords: Emotion Recognition, Personalized Recommendations, Computer Vision, Deep Learning, Convolutional Neural Networks (CNN), Fashion Technology

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